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Attention Is All You Need

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The dominant sequence transduction models are based on complex recurrent or convolutional neural networks in an encoder-decoder configuration. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

Comments: 15 pages, 5 figures

Subjects: **Computation and Language (cs.CL)**; Machine Learning (cs.LG)

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Attention Is All You Need

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Introduction

Background and Motivation

- RNN, LSTM, GRN which have encoder-decoder architectures have been established SOTA approaches in sequence modeling and transduction problems.
- Recurrent models typically factor computation along the symbol positions of the input and output sequences. This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths, as memory constraints limit batching across examples.
- Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences.
- Transformer, a model architecture eschewing recurrence and instead relying entirely on an **attention mechanism** to draw global dependencies between input and output.

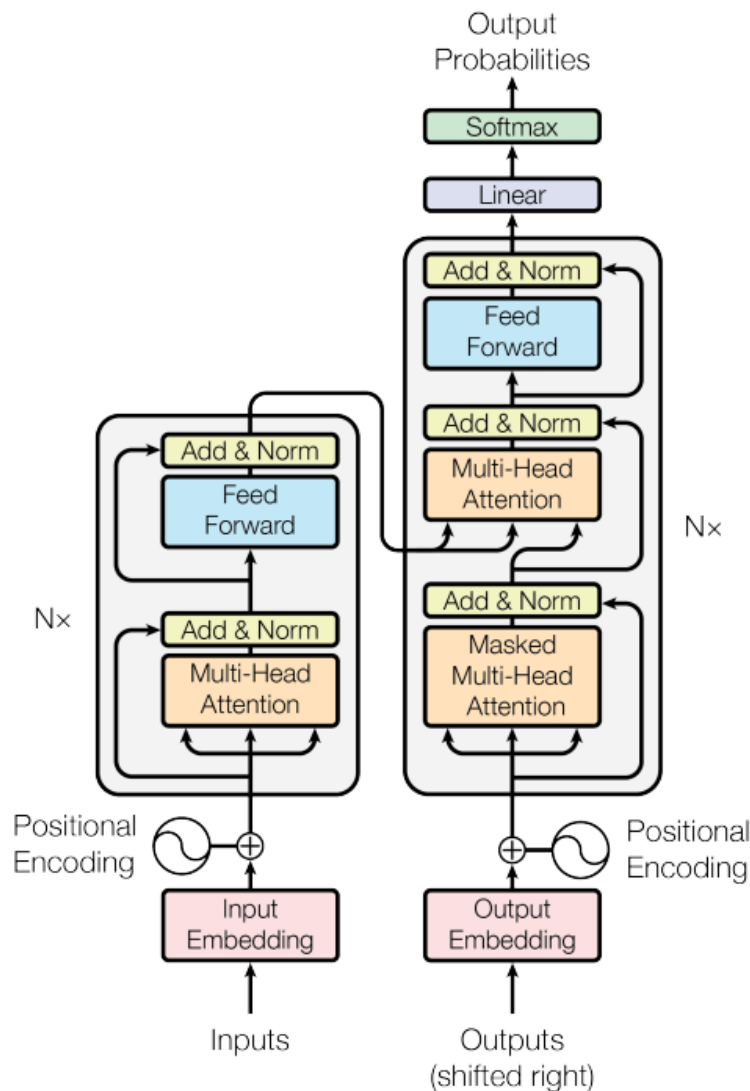
Background

Data acquisition and preprocessing

- ConvS2S and ByteNet are based on CNNs, enabling parallel computation.
- However, they struggle to learn dependencies between distant positions.
- The Transformer leverages self-attention to model such dependencies with a constant number of operations, regardless of distance.
- It is the first model to generate input and output representations without relying on RNNs or CNNs.
- The Transformer addresses the limitations of self-attention through Multi-Head Attention.

Model Architecture

Encoder and Decoder Stacks



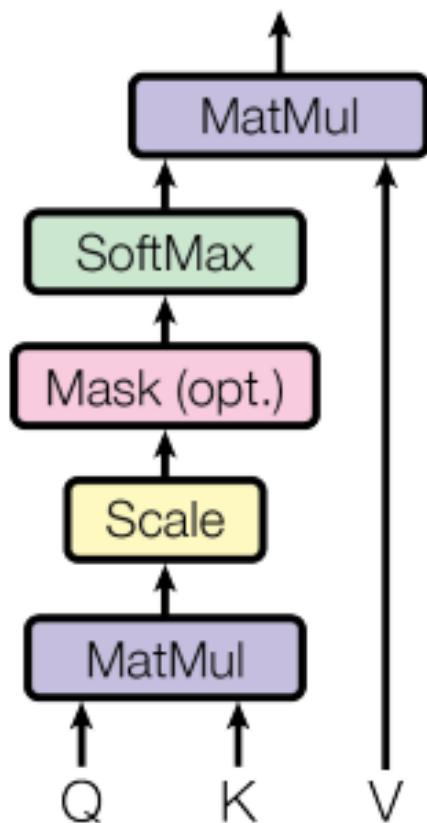
- **Encoder:**
 - $N=6$ (each has 2 sub-layer)
 - MHA mechanism
 - simple, position wise FFN
 - residual connection around each of the two sub-layers, followed by layer normalization
- **Decoder:**
 - $N=6$ (each has 3 sub-layer)
 - 3rd layer performs performs MHA over the output of the encoder stack
 - Masked MHA ensures that the predictions for position i can depend only on the known outputs at positions less than i

Figure 1: The Transformer - model architecture.

Model Architecture

Scaled Dot Product Attention

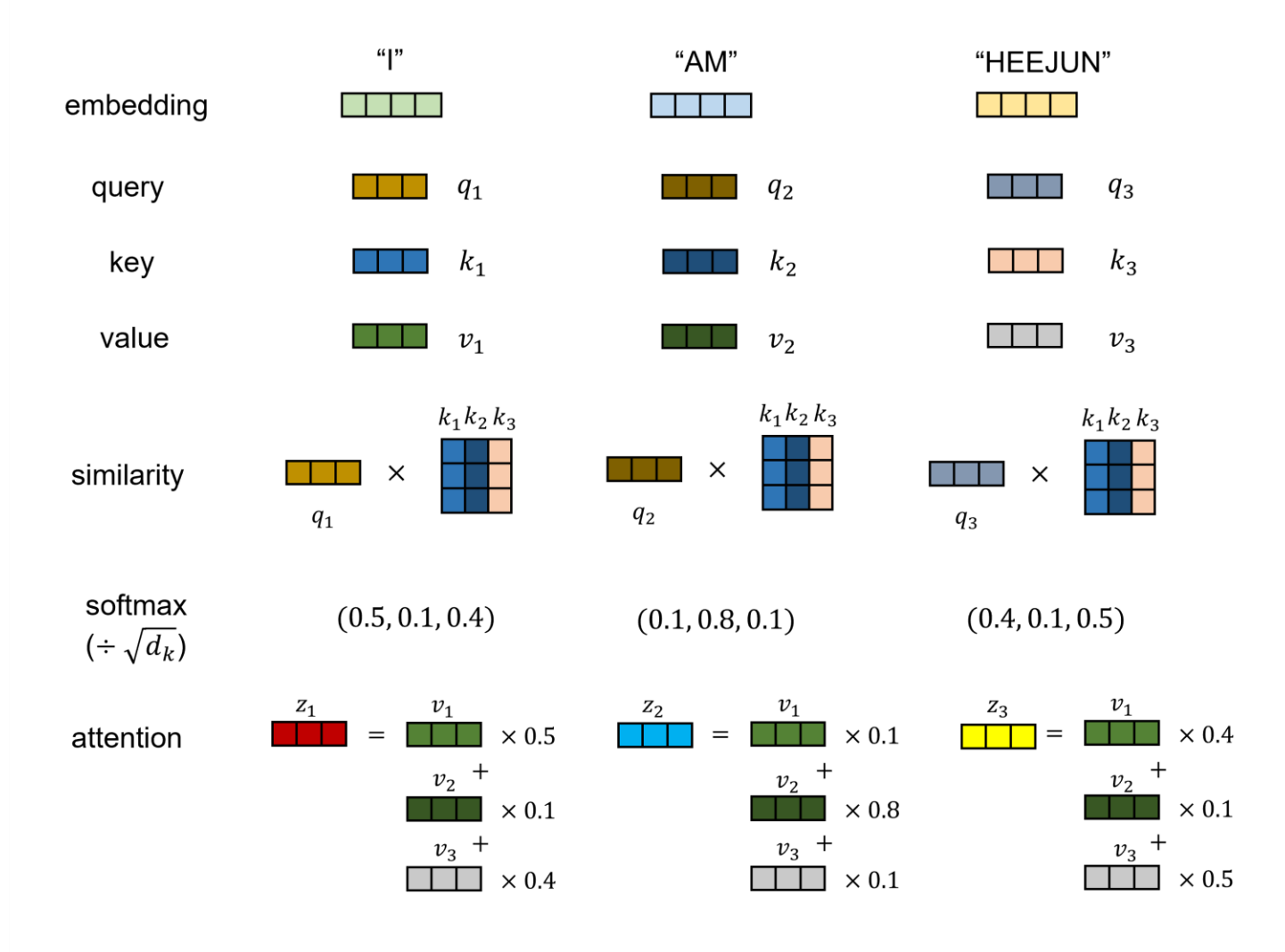
Scaled Dot-Product Attention $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$



- **Q, K**: dimension dk
- **V**: dimension dv Compute dot product of Q and K
- Scale by root dk
- Apply softmax and multiply by V
 - Without scaling, large dot products can push softmax into small gradient regions
 - Leads to slow or unstable training
 - Scaling by dk fixes this

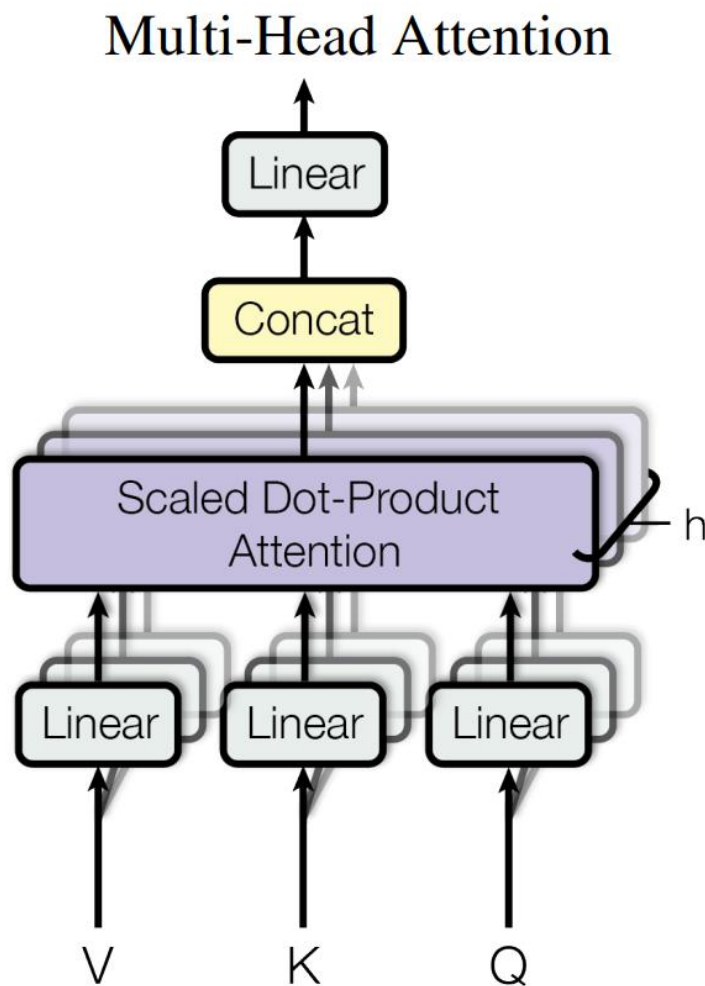
Model Architecture

Scared Dot Product Attention



Model Architecture

Multi-Head Attention



- Instead of using a single attention function, MHA runs multiple attention heads in parallel.
- Each head uses different learned linear projections of the input Q, K, and V.
- a 512-dimensional input is projected into 8 separate 64-dimensional Q/K/V sets.
- Each head performs Scaled Dot-Product Attention independently.
- The outputs are concatenated and passed through a final linear layer.
- This allows the model to attend to information from different representation subspaces at different positions.

Model Architecture

Multi-Head Attention

Which do you like better, coffee or tea?

- 문장 타입에 집중하는 어텐션

Which do you like better, coffee or tea?

- 명사에 집중하는 어텐션

Which do you like better, coffee or tea?

- 관계에 집중하는 어텐션

Which do you like better, coffee or tea?

- 강조에 집중하는 어텐션

Why Self-Attention

VS CNN, RNN ...

- 1st Total computational complexity per layer.
- 2nd the amount of computation that can be parallelized.
- 3rd the path length between long-range dependencies in the network

Layer Type	Complexity per Layer	Sequential Operations	Maximum Path Length
Self-Attention	$O(n^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent	$O(n \cdot d^2)$	$O(n)$	$O(n)$
Convolutional	$O(k \cdot n \cdot d^2)$	$O(1)$	$O(\log_k(n))$
Self-Attention (restricted)	$O(r \cdot n \cdot d)$	$O(1)$	$O(n/r)$

Why Self-Attention

Side benefit

- self-attention could yield more interpretable models.

